

assignment equilibrium

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:34pm

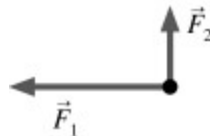
Quiz Instructions

There is a tutorial



Question 1 6.7 pts

The figure shows two forces acting at right angles on an object. They have magnitudes $F_1 = 6.3$ N and $F_2 = 2.1$ N. What third force will cause the object to be in equilibrium (acceleration equals zero)?



4.2 N at 162° counterclockwise from $\vec{F}_1$



6.6 N at 108° counterclockwise from $\vec{F}_1$



4.2 N at 108° counterclockwise from $\vec{F}_1$

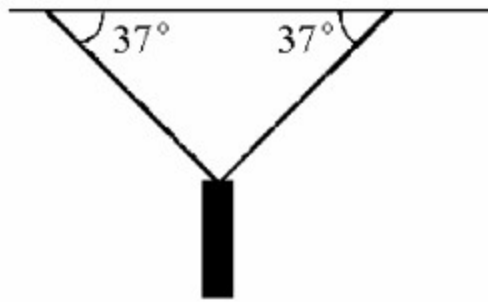


6.6 N at 162° counterclockwise from $\vec{F}_1$



Question 2 6.7 pts

A traffic light weighing 100 N is supported by two ropes as shown in the figure. The tensions in the ropes are closest to


☐

83 N.

☐

66 N.

☐

63 N.

☐

56 N.

☐

50 N.



Question 3 6.7 pts

A stalled car is being pushed up a hill at constant velocity by three people. The net force on the car is

☐

down the hill and greater than the weight of the car.

☐

up the hill and equal to the weight of the car.

☐

down the hill and equal to the weight of the car.

☐

zero.

☐

up the hill and greater than the weight of the car.



Question 4 6.7 pts

Figure 5.4

Dick and Jane stand on a platform of negligible weight, as shown in Figure 5.4. Dick weighs 500 N and Jane weighs 400 N. Jane is supporting some of her weight on the end of the rope she is holding. What is the downward force she is exerting on the platform? watch tutorial for help.

☐

50 N

☐

100 N

☐

0

☐

240 N

☐

300 N



Question 5 6.7 pts

A satellite is in orbit around the earth. Which one feels the greater force?

☐

It depends on the distance of the satellite from Earth.

☐

the earth because the satellite has so little mass

☐

Earth and the satellite feel exactly the same force.

☐

the satellite because the earth is so much more massive



Question 6 6.7 pts

An object of weight W is in freefall close to the surface of Earth. The magnitude of the force that the object exerts on Earth is

☐

zero.

☐

less than W .

☐

cannot be determined without knowing the relative masses of the object and the earth.

☐

greater than W .

☐

equal to W .



Question 7 6.7 pts

Figure 5.10 shows a 4,800 kg cable car descending a high hill. A counterweight of mass 4,600 kg on the other side of the hill aids the breaks in controlling the cable car's speed. The rolling friction of both the cable car and the counterweight are negligible. How much braking force does the cable car need to descend at constant speed?

Figure 5.10

☐

8,100 N

☐

980 N

☐

5,800 N



3,500 N



Question 8 7 pts

In a collision between a huge SUV and a small hybrid car, the SUV exerts a larger force on the hybrid than the hybrid exerts on the SUV.



True



It depends on whether the collision is a head-on collision or a rear-end collision.



False



Question 9 6.7 pts

A 10-kg sign is held by two ropes as shown in the figure. What is the tension on rope A?



44 N



69 N



72 N



98 N



88 N



Question 10 6.7 pts

Figure 5.6

A 6.0 kg box is held at rest by two ropes that form 30° angles with the vertical. An external force F acts vertically downward on the box. The force exerted by each of the two ropes is denoted by T . A force diagram, showing the four forces that act on the box in equilibrium, is shown in Figure 5.6. The magnitude of force F is 410 N. The magnitude of force T is closest to:

☐

271 N

☐

376 N

☐

235 N

☐

188 N

☐

470 N

☐

Question 11 6.7 pts

Two weights are connected by a massless wire and pulled upward with a constant speed of 1.50 m/s by a vertical pull P . The tension in the wire is T (see figure). Which one of the following relationships between T and P must be true?

☐

$$P + T = 125 \text{ N}$$

☐

$$P = T + 100 \text{ N}$$

☐

$$T > P$$

☐

$$P = T + 25 \text{ N}$$

☐

$$T = P$$



Question 12 6.7 pts

In the figure, a block of mass M hangs at rest. The rope that is fastened to the wall is horizontal and has a tension of 52 N. The rope that is fastened to the ceiling has a tension of T and makes an angle θ with the ceiling. What is the angle θ ?

- ☐ 55°
- ☐ 30°
- ☐ 35°
- ☐ 63°
- ☐ 45°
- ☒

Question 13 6.7 pts

The figure shows two forces, each of magnitude 4.6 N, acting on an object. The angle between these forces is 40° , and they make equal angles above and below the horizontal. What third force will cause the object to be in equilibrium (acceleration equals zero)?

- ☐ 7.0 N pointing to the right
- ☐ 3.5 N pointing to the right



8.6 N pointing to the right



4.3 N pointing to the right



Question 14 6.7 pts

Two unequal masses M and m ($M > m$) are connected by a light cord passing over a pulley of negligible mass, as shown in the figure. When released, the system accelerates. Friction is negligible. Which figure below gives the correct free-body force diagrams for the two masses in the moving system?





Question 15 7 pts

A brick is resting on a rough incline as shown in the figure. The friction force acting on the brick, along the incline, is

☐

equal to the weight of the brick.

☐

greater than the weight of the brick.

☐

less than the weight of the brick.

☐

zero.

Quiz saved at 7:35pm

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